



Final Year Project

Vision & Scope Document

Project Members

Name	Roll Number
M. Sohaib Saleem (Lead)	2K22 BSCS 301
Abdul Wahab	2K22 BSCS 330
Muhammad Rehan	2K22 BSCS 339
Abdullah Khan	2K22 BSCS 342

Table of Contents

Vision and Scope Document.....	1
1. Business Requirements & Background.....	1
1.1 The Problem.....	1
1.2 Why This Product Is Needed.....	1
2. Product Vision.....	1
3. Scope of Initial Release (Version 1.0).....	2
4. Product Roadmap (Post-Launch).....	2
Release 1.1: The "Customization" Update.....	2
Release 2.0: The "Expanded Intelligence" Update.....	2
5. Future Assumptions & Dependencies.....	3

Vision and Scope Document

Project: Intelligent, State-Aware Security and Monitoring System

Author: Muhammad Sohaib Saleem [Product Lead / CEO]

Date: December 25, 2025

1. Business Requirements & Background

1.1 The Problem

The current home security market is saturated with "smart" devices that are often unintelligent and intrusive.

- **False Alarm Fatigue:** Existing commercial solutions suffer from a high frequency of false alarms caused by non-threats like pets, weather, or passing vehicles. This leads to user fatigue and a loss of trust in the system's reliability.
- **Lack of Control:** Most systems operate in a static, "always-on" mode. Users lack the ability to easily toggle privacy settings when they are home, resulting in a cumbersome user experience.
- **Cost & Privacy:** High-end systems often come with expensive subscription costs and privacy concerns regarding where video data is processed.

1.2 Why This Product Is Needed

Our product addresses these shortcomings by introducing "**State-Aware Logic.**" We are not just building a camera; we are building a system that understands context. By integrating user-driven modes (Active/Inactive) with on-device Machine Learning, we provide a solution that is:

- **Intelligent:** It filters out false positives by distinguishing between genuine human threats and harmless motion.
- **Respectful:** It offers absolute privacy through an "Inactive Mode" that stops all recording when the user is home.
- **Efficient:** It captures high-value evidence (video + interval stills) only when necessary, optimizing storage and bandwidth.

2. Product Vision

For homeowners and small business owners who need a reliable, low-maintenance security solution. The Intelligent State-Aware System is a smart monitoring platform that eliminates false alarms and puts privacy control in the hands of the user. Unlike standard motion-activated cameras like Ring or Nest10, our Product uses edge computing

(Raspberry Pi) to perform AI verification locally before ever alerting the user.

3. Scope of Initial Release (Version 1.0)

This release constitutes the core Final Year Project submission.

The initial release focuses on establishing the core "State-Aware" ecosystem.

- **Hardware:** Raspberry Pi 4 unit with Camera Module and PIR Sensor.
- **Core Logic:**
 - **PIR Wake-up:** System sleeps until motion is detected.
 - **AI Verification:** TensorFlow Lite model (SSD MobileNet V2) confirms human presence.
 - **Intrusion Protocol:** Records 1-minute video + captures stills every 30 seconds.
- **Software Interface:**
 - **Web Dashboard:** Responsive app for toggling Active/Inactive modes.
 - **Real-Time Alerts:** Instant notifications via Socket.IO.
- **Data Management:** Secure local storage with MongoDB and GridFS for handling video evidence.

4. Product Roadmap (Post-Launch)

As we look beyond the initial prototype, we have structured a roadmap for future development to transform this from a student project into a commercial-grade product.

Release 1.1: The "Customization" Update

- **Objective:** Give users granular control over system behavior.
- **Features:**
 - **Configurable Logic:** Allow users to set "Intrusion Mode" duration and image capture intervals via the dashboard.
 - **Sensitivity Tuning:** Software controls to adjust motion sensor sensitivity remotely.

Release 2.0: The "Expanded Intelligence" Update

- **Objective:** broaden the scope of detection and accessibility.
- **Features:**

- **Advanced AI Models:** Upgrade the ML model to detect specific objects beyond humans, such as vehicles or delivery packages.
- **Mobile Ecosystem:** Native iOS and Android applications for push notifications and background operation.
- **Multi-User Support:** Introduction of Admin and Guest roles, allowing families or teams to monitor a single location.

5. Future Assumptions & Dependencies

- **Environmental Expansion:** We assume a future demand for holistic home monitoring. The architecture is designed to eventually support environmental sensors (temperature, humidity, air quality) alongside security data.
- **Data Visualization Trends:** We anticipate users will want to see patterns in their security data. Future versions will need to include data visualization tools (graphs/charts) to track activity trends over time.
- **Hardware Availability:** The project assumes the continued availability and support of the Raspberry Pi ecosystem and compatible TensorFlow Lite libraries.